

DEXTER FERNANDES

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SUMMARY

Computer Vision and Perception Engineer with 3+ years building real-time vision pipelines on edge hardware, from camera acquisition through detection, tracking and field deployment. Built a GStreamer and TensorRT pipeline with a custom CUDA plugin, and a ROS2 SLAM stack on an NVIDIA Jetson AGX Xavier fusing LiDAR and camera sensing. Skilled in OpenCV-based detection and tracking, camera calibration and 3D geometry, and 24/7 production perception under low-latency constraints. MSc Robotics (Merit), University of Bristol.

RELEVANT EXPERIENCE

Esbaar — AI Engineer

01/2023 – 08/2023 · MUSCAT, OMAN

- Developed and maintained real-time NVIDIA DeepStream and GStreamer video pipelines processing high-resolution, high-FPS camera feeds for 24/7 perception on BP drilling rigs, covering ingestion, inference and field diagnostics in desert conditions.
- Owned the end-to-end ML lifecycle for real-time object detection on BP rigs, adapting YOLOv5 detectors in PyTorch and benchmarking against EfficientDet on a repeatable harness, improving F1 by 15% under 24/7 operation in three months.
- Applied OpenCV-based image preprocessing and artefact detection to catch mud and sand contamination on camera feeds, cutting false positives by 13%.
- Hardened deployed services with Dockerised microservices and a watchdog application with a REST API, cutting live video pipeline crashes by 67% and enabling remote recovery without site visits.
- Led on-site validation on BP rig deployments, diagnosing perception failures under lighting, motion and environmental stressors and turning findings into fixes for live operation.
- Generated synthetic images and augmentation routines with OpenCV and scikit-image to cover edge cases in harsh industrial conditions, improving robustness on contaminated feeds.

LivNSense Technologies — Lead Software Engineer, AI

05/2020 – 12/2022 · BENGALURU, INDIA

- Led four engineers building a two-stage YOLOv5-based detector with helmet-colour classification for construction safety, cutting false positives by 33% within latency and power budgets on Intel edge hardware.
- Built asynchronous C++17 perception pipelines on OpenVINO across Intel Movidius VPUs and industrial edge boxes, overlapping preprocessing, inference and post-processing to raise throughput from 5 to 15 FPS under real-time constraints.
- Built a real-time multi-camera perception pipeline on a moving construction vehicle, synchronising streams and tuning camera and ISP settings across RGB and IR feeds to cut false positives on IR and reduce manual recalibration.
- Added ByteTrack multi-object tracking over detections to hold worker and vehicle identities across frames, cutting duplicate incident alerts from the moving-vehicle feeds.
- Set experimentation standards, reviewed model performance and mentored junior engineers, guiding optimisation strategy for production deployments.
- Evaluated DeepLabv3+, Mask R-CNN, U-Net and SegFormer for annotation review and failure analysis on industrial camera feeds.

PROJECTS

DeepStream RTSP Pipeline github.com/Dexter-Fernandes/deepstream-rtsp-pipeline

06/2026

- Developed and evaluated an end-to-end DeepStream and GStreamer RTSP pipeline running YOLO detection and NvDCF tracking across 3 concurrent feeds, with metadata extraction and re-streamed output.
- Implemented a C++ and CUDA TensorRT decode plugin with IPluginV2DynamicExt, moving post-processing into the inference engine, and converted YOLO26n to a dynamic-batch TensorRT FP16 engine sustaining 29.7 FPS per stream.
- Designed a geometry-first cross-camera (MTMC) identity layer with uncertainty-aware ground-plane homographies per camera, lifting pooled IDF1 from 0.198 to 0.245 on the WildTrack benchmark.
- Hardened the pipeline for realistic deployment conditions with NTP-based timestamps across RTSP sources, clock-skew gating and structured health monitoring to flag stalled streams in real time.
- Added an optional DeepStream ReID SGIE for sparse appearance fusion; a 28% throughput hit for only 0.002 F1 gain over geometry-only led to shipping geometry as the default tracker.

ROS2 SLAM and Nav2 Simulation

02/2026

- Developed and evaluated a simulation-based robotics stack in ROS2 Jazzy and Gazebo Harmonic, comparing LiDAR-centric SLAM, LiDAR-camera fusion SLAM and autonomous navigation in structured scenarios.
- Built ROS2 nodes, launch files and parameterised configs in C++ and Python, using tf2 and message synchronisation to debug transform timing issues.
- Implemented point cloud filtering, segmentation and registration with PCL and Open3D, and added an EKF localisation layer fusing wheel odometry and IMU.
- Containerised the stack with Docker on Linux and managed versioning in Git for reproducible builds across development environments.

Modern Edge CV Benchmarking github.com/Dexter-Fernandes/ModernEdge-CV-Benchmarking

03/2026

- Benchmarked modern PyTorch detectors including YOLOv8, YOLOv10, YOLO26 and RT-DETR on accuracy, latency and model size to assess deployment suitability for edge targets.
- Exported models to ONNX and evaluated OpenVINO deployment paths, comparing mAP, inference time and model size to support model selection for low-latency systems.

- Extended the comparison to EfficientViT, MobileSAM and Depth Anything v2, evaluating Vision Transformer and foundation-model approaches across detection, segmentation and depth estimation.

EDUCATION

University of Bristol — MSc Robotics, Merit 09/2023 – 09/2024 · BRISTOL, ENGLAND
Dissertation: Radiological SLAM with LiDAR Odometry

- Built a real-time SLAM system in C++17 and ROS2 (roslcpp) on a Clearpath Husky, fusing Velodyne VLP-16 LiDAR and RealSense D435i depth for inspection mapping, run on an NVIDIA Jetson AGX Xavier under JetPack.
- Calibrated LiDAR extrinsics to the robot base and resolved transform timing with tf2 across a PCL point cloud pipeline, validating sensor placement for coverage.
- Benchmarked RTAB-Map, LIO-SAM and Cartographer across matched rosbag2 sequences under a single evaluation protocol to compare mapping and localisation behaviour.
- Configured Nav2 in simulation for autonomous waypoint traversal, testing survey routes before running them on the Husky.

Visvesvaraya National Institute of Technology — BTech Electrical and Electronics Engineering 07/2016 – 05/2020 · NAGPUR, INDIA
▪ Foundation in signals, systems, embedded hardware and control, later applied to real-time perception, sensor integration and robotics software.

SKILLS

Languages	Python, C++17, CUDA C++, Bash
Computer Vision	object detection, object tracking, image classification, semantic and instance segmentation, optical flow, YOLO (v5/v8/v10/v26), RT-DETR, OpenCV, PyTorch, TensorFlow, NumPy
Video Pipelines	NVIDIA DeepStream, GStreamer, FFmpeg, RTSP, nvinfer, nvstreammux, nvtracker, NvDCF, NvSORT, ByteTrack, multi-camera synchronisation, multi-target multi-camera (MTMC) tracking, ISP and camera tuning (RGB and IR)
Edge Inference	NVIDIA Jetson AGX Xavier, JetPack, TensorRT (custom plugins, dynamic batching), OpenVINO, ONNX, CUDA, quantisation-aware training, FP16/INT8, structured pruning, layer fusion, latency profiling, Intel Movidius VPU
Robotics and SLAM	ROS2 (Robot Operating System 2), roslcpp, rclpy, tf2, rosbag2, Gazebo, RTAB-Map, LIO-SAM, Cartographer, Nav2, LiDAR SLAM, ICP (Iterative Closest Point), LiDAR extrinsics calibration, PCL, Open3D, EKF
3D Geometry	camera calibration, ground-plane homography, 3D coordinate transformations, sensor extrinsics calibration, multi-camera calibration
Embedded and Production	embedded Linux, Docker, CI/CD, GitHub Actions, Git, CMake, pytest, gtest, REST API design, structured logging, health monitoring, observability
MLOps and Data	synthetic data generation, benchmarking harnesses, ablation studies, Weights & Biases, dataset versioning, COCO annotation format
Professional	technical leadership, mentoring, Agile and Scrum, stakeholder communication

ADDITIONAL EXPERIENCE

GrowthStage — Technical Assessor, Computer Vision and Software Engineering 05/2026 – PRESENT · REMOTE
Assesses candidate technical depth across perception, SLAM and deployment using hands-on robotics and computer vision knowledge.

CERTIFICATIONS AND AWARDS

Deep Learning Specialisation, deeplearning.ai (07/2019) Best Speaker award, ABC Toastmasters Club (01/2020)